

SWEN 670 — Software Engineering Capstone

CareConnect Application

Team B — Project Plan

Summer 2026 | University of Maryland Global Campus

Professor Mir Assadullah

Document Version 1.4

Role	Name	Responsibilities
Team Lead	Alex Yom	Project coordination, milestone planning, 661 integration
Member	Brice Tikum	Test Plan, Coverage Improvement Plan
Member	Prashanth Saseenthara	Testing Baseline, SRS testing requirements, SRS user classes (Section 2.2)
Member	Lekan Ogunsanya	SRS deployment & infrastructure, SRS use cases (Section 6.1)

Version	Date	Description
1.0	5/13/2026	Initial draft
1.1	5/28/2026	Removed Rashaad Bell (withdrew from course); redistributed SRS user classes (Section 2.2) to Prashanth Saseenthara and SRS use cases (Section 6.1) to Lekan Ogunsanya
1.2	5/29/2026	Updated team structure, roles table, and resource cost estimation section per class standard (Joriel Rivas, Version 1.0); incorporated WBS, Budget Summary, and EVM tracking tables

1.3	5/30/2026	Updated Budget Summary and WBS per professor feedback: increased Technical Architect/DevOps hours from 25 to 50 to reflect CloudFormation CI/CD pipeline build; added Team Lead/PM hours (40); corrected budget totals to \$33,130 across 655 hours; updated Open Questions to reflect resolved items
1.4	6/7/2026	

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Executive Summary

Team B is responsible for ensuring the quality, reliability, and accessibility of the CareConnect application as part of the SWEN 670 Software Engineering Capstone, Summer 2026 at the University of Maryland Global Campus. CareConnect is a full-stack healthcare application connecting caregivers and patients through a Flutter/Dart frontend and Spring Boot Java backend.

Team B's scope focuses on four core areas: comprehensive unit and end-to-end testing infrastructure, achieving and maintaining tiered code coverage targets across all modules — 100% on high-risk transactional logic (Shift Scheduling, EVV) and 90% on visual/placement components — with full VPAT conformance statement by Milestone 4. Team B's work is foundational to the entire class — all teams depend on Team B to validate that delivered features meet quality and accessibility standards before integration.

This Project Plan defines Team B's team structure, milestone deliverables, project schedule, Git branching strategy, SWEN 661 coordination process, resource and cost estimation, risk management, and open questions requiring resolution. The plan is structured to align with the four course milestones: Project Initiation (May 30), Design and Prototyping (June 13), Development (July 25), and Final Delivery (August 4, 2026).

Team B consists of four members following the withdrawal of one team member in Week 2. All members have reviewed and signed this Project Plan in acknowledgment of their assigned responsibilities.

1. Introduction

1.1 Purpose

This Project Plan defines the scope, schedule, deliverables, and team structure for Team B's contributions to the CareConnect application during the SWEN 670 Software Engineering Capstone, Summer 2026. Team B is responsible for comprehensive testing, WCAG and Section 508 accessibility compliance, and production of a Voluntary Product Accessibility Template (VPAT).

1.2 Project Overview

CareConnect is a full-stack healthcare application designed to streamline communication and coordination between caregivers and patients. It includes a Flutter/Dart frontend and a Spring Boot Java backend. Team B's work supports the entire class by ensuring the application meets quality and accessibility standards across all milestones.

1.3 Scope

Team B's responsibilities span the full quality and accessibility lifecycle of the CareConnect application. The following areas define the boundaries of Team B's work across all four milestones:

- Unit testing — increasing and maintaining unit test coverage across all modules
- End-to-end (E2E) testing — validating critical user workflows across the integrated application
- Tiered code coverage — 100% on high-risk transactional modules (Shift Scheduling, EVV) and 90% on visual/placement components, with automated reporting via CloudFormation-managed CI/CD pipeline
- WCAG 2.1 AA compliance — auditing and documenting accessibility across the application
- Section 508 compliance — evaluating functional accessibility criteria
- VPAT production — compiling accessibility findings into a formal VPAT document
- 661 integration validation — reviewing and testing UI components delivered by SWEN 661

1.4 Out of Scope

To maintain focus and avoid overlap with other teams, the following areas fall outside Team B's responsibilities. These items are owned by other teams within SWEN 670 and are listed here to provide clarity on boundaries:

- Feature development (owned by Teams A, C, D, E)
- Backend infrastructure deployment (Team A)
- UI design and multilingual implementation (Team C)
- EVV features, invoicing, wearable device integration (Team D)
- USPS integration, AI agent/call summaries (Team E)

2. Team Structure and Roles

Team B consists of four members with complementary skill sets. All members are responsible for reviewing each other's work before submission. No section is considered complete until the full team has reviewed it.

Name	Background	Primary Milestone 1 Assignment
Alex Yom	Team Lead	Project Plan overview, team roles, schedule, 661 coordination
Brice Tikum	Software Engineer/Analyst	Test Plan + Coverage Improvement Plan
Prashanth Saseenthara	Configuration & Testing Specialist	Testing Baseline document SRS — Use Classes
Lekan Ogunsanya	Systems Administrator / AWS	SRS — User Requirements, Deployment & Infrastructure

3. Milestone Plan

The course runs from May 13, 2026 to August 4, 2026 with four milestones. Team B deliverables for each milestone are defined below. All milestone deliverables must be reviewed by the full team before submission.

3.1 Milestone 1 — Project Initiation (Due: May 30, 2026)

Milestone 1 focuses on project initiation, establishing Team B's planning foundation, and documenting the current state of testing and accessibility in the CareConnect codebase. The following deliverables are required from Team B by May 30, 2026:

- Project Plan (this document) — Team B section
- Software Requirements Specification (SRS) — Team B sections
- Testing Baseline — current framework, coverage %, existing tests, identified gaps
- Test Plan — unit and E2E testing strategy
- Coverage Improvement Plan — roadmap to 95%+ coverage
- VPAT/Accessibility Plan — approach, tools, and timeline
- Individual submissions — screenshot of CareConnect running locally (all members)

3.2 Milestone 2 — Design and Prototyping (Due: June 13, 2026)

Milestone 2 transitions Team B from planning into early execution. All updated documents must use Word Track Changes to reflect revisions from Milestone 1. The following deliverables are required from Team B by June 13, 2026:

- Updated Project Plan and SRS (tracked changes)
- Technical Design Document — testing architecture and tooling
- Software Test Plan — detailed test cases and coverage targets
- Initial unit test implementation — high-priority components
- Initial E2E test implementation — critical user flows
- Early accessibility review findings

3.3 Milestone 3 — Development (Due: July 25, 2026)

Milestone 3 represents the core development and testing phase of the project. Team B is expected to demonstrate measurable progress toward the 95%+ coverage target and provide substantive accessibility evidence. All updated documents must include tracked changes. The following deliverables are required from Team B by July 25, 2026:

- Updated documents (tracked changes)
- Programmer Guide — testing conventions and setup

- Expanded test coverage — approaching 95%+ target
- Regression testing on team-b-develop after Monday integration merges
- Continued accessibility and 508 compliance review
- Updated VPAT-related documentation

3.4 Milestone 4 — Final Delivery (Due: August 4, 2026)

Milestone 4 is the final delivery milestone. All work must be complete, documented, and demo-ready. Team B's primary deliverables center on the finalized test report, verified coverage metrics, and the completed VPAT. The following deliverables are required from Team B by August 4, 2026:

- Final test coverage report (95%+ verified)
- Final accessibility evidence and remediation documentation
- Completed VPAT document
- User Guide — testing and quality sections
- Test Report — full summary of all testing activities
- Final quality review and demo evidence

4. Project Schedule

Milestone	Due Date	Key Team B Deliverables	Status
Milestone 1	May 30, 2026	Project Plan, SRS, Testing Baseline, Test Plan	Complete
Milestone 2	June 13, 2026	Tech Design Doc, Software Test Plan, Initial Tests	<i>[Not Started]</i>
Milestone 3	July 25, 2026	Expanded Coverage, Accessibility Review, VPAT Draft	<i>[Not Started]</i>
Milestone 4	August 4, 2026	Final Test Report, VPAT, Coverage 95%+	<i>[Not Started]</i>

Weekly recurring events:

- Monday & Thursday 6:00pm ET — Team Lead meetings (Alex attends)
- Monday Integration Meeting — team-b-develop reviewed and merged to develop
- End of each week — peer evaluations due via Google Form

5. Git Branching Strategy

Team B follows the class-wide SWEN 670 CareConnect Git Branch Policy established by the class PM (Joriel Rivas).

5.1 Branch Hierarchy

The CareConnect repository follows a structured multi-tier branching model established by the class PM, Joriel Rivas. Each branch serves a distinct purpose in the integration and release flow, ensuring that code is validated at every level before reaching the production-ready main branch. The hierarchy is as follows:

- main — production ready, leads only, weekly merge
- develop — shared integration branch, merges from team branches on Mondays
- team-b-develop — Team B's integration branch, owned by Alex
- feature/b-* — individual feature branches, one per task

5.2 Daily Workflow

To maintain a clean and conflict-free codebase, all Team B members are expected to follow this workflow for every task without exception. This sequence ensures that each member is always working from the latest version of the team branch and that all changes are properly reviewed before integration:

- git checkout team-b-develop && git pull origin team-b-develop
- git checkout -b feature/b-your-task-name
- Complete work, then: git add . && git commit -m "feat: description"
- git push -u origin feature/b-your-task-name
- Open a Pull Request targeting team-b-develop

5.3 Definition of Done

A feature branch is not considered ready for merge until all of the following criteria have been met. This definition applies to every pull request targeting team-b-develop, regardless of task size or urgency:

- Code is complete
- Unit tests written and passing
- All tests passing
- No merge conflicts
- Build successful
- README updated if needed

6. SWEN 661 Coordination

SWEN 661 serves as the consulting firm for the CareConnect project, while SWEN 670 acts as the client. SWEN 661 is responsible for designing and delivering UI components that are then handed off to the 670 teams for validation, integration, and testing. This relationship mirrors a real-world client-vendor engagement and requires clear communication, defined handoff criteria, and structured review processes.

6.1 Point of Contact

Alex Yom (Team B Lead) serves as the primary point of contact between Team B and the SWEN 661 class. All UI component deliveries, accessibility feedback, and integration communications are routed through Alex. This single point of contact ensures accountability, reduces miscommunication, and creates a clear audit trail for all cross-class coordination.

6.2 Handoff and Validation Process

To ensure quality and accessibility standards are upheld before any SWEN 661 component is integrated into the 670 codebase, Team B follows a structured five-step validation process for every component delivery. No component may be merged without completing each step in sequence:

- Alex receives the component delivery and reviews it against the agreed requirements
- Team B runs unit tests and accessibility checks against the delivered component
- Any accessibility or quality issues are documented and communicated back to 661 via Alex before integration proceeds
- Once validated, the component is merged into the team-b-develop branch following the class Git branching policy
- WCAG and Section 508 findings related to 661 components are documented for inclusion in the VPAT

6.3 Grading and Submission Boundaries

It is important to note that SWEN 661 and SWEN 670 are separate courses with independent grading. Team B will not submit 661 work as part of SWEN 670 deliverables, and vice versa. Collaboration is encouraged, but each course's submissions must reflect that course's own work. Any UI components received from 661 are treated as external deliverables to be validated and integrated — not co-authored.

6.4 Communication Cadence

All coordination between Team B and SWEN 661 follows a structured communication cadence to ensure transparency, timely feedback, and a clear audit trail. The following practices govern how and when cross-class communication occurs:

- Alex attends Monday/Thursday 6pm ET team lead meetings where cross-team coordination is discussed

- Any 661 deliverables received mid-week are reviewed by Team B before the following Monday integration meeting
- Slack is used as the primary communication channel for all coordination between Team B and 661
- Professor Assadullah is included as an optional attendee on cross-class meetings as needed

7. Resource and Cost Estimation

Risk management is an ongoing process throughout the project lifecycle. The following risk register documents all identified risks, their likelihood and impact ratings, and the mitigation strategies Team B will employ. Each risk is color-coded according to the risk matrix defined above — Red indicating high priority requiring immediate attention, Yellow indicating moderate priority requiring monitoring, and Green indicating low priority. The register will be reviewed and updated at each milestone.

Standard Budget Role Categories and Hourly Rates

Role	Use This for	BLS Occupation Used for Rate	Hourly Rate
Team Lead / PM	Scheduling, team coordination, meeting preparation, EVM tracking, risk tracking, milestone coordination, presentation coordination	Project Management Specialist	\$48
Business Analyst/Requirements Analyst	Requirements gathering, scope clarification, stakeholder analysis, business rules, use-case support, project planning input	Management Analyst	\$49
UI/UX Designer	Interface design, usability, responsive design, accessibility layout design, prototype or mockup support	Web and Digital Interface Designer	\$47
Technical Architect / DevOps / Cloud Architect	System architecture, AWS/cloud planning, deployment planning, integration planning, technical direction, infrastructure decisions	Computer Network Architect	\$63
Software Engineer / Developer / Programmer	Coding, implementation, feature development, integrations, multilingual/voice/AI feature work, bug fixes	Software Developer	\$64
QA / Tester / Accessibility Tester	Unit testing, E2E testing, test planning, defect validation, WCAG/508 testing, regression testing, test reports	Software Quality Assurance Analyst / Tester	\$49
Technical Writer / Documentation Specialist	Project Plan drafting, technical documentation, user guide, programmer guide, deployment guide, final documentation cleanup.	Technical Writer	\$48

WBS Budget

The Work Breakdown Structure (WBS) below decomposes Team B's full scope of work into individual tasks and work packages mapped to specific milestones, role categories, estimated hours, and costs. Each WBS item corresponds to a deliverable that will be tracked through the EVM reporting table:

WBS ID	Task / Work Package	Deliverable	Milestone	Budget Role Category	Estimated Hours	Hourly Rate	Estimated Cost
1.0	Team Lead Coordination & Milestone Management	Project oversight, meeting prep, EVM tracking, presentations	M1-M4	Team Lead / PM	40	\$48	\$1920
1.1	Project Plan Development	Project Plan (Team B)	M1	Technical Writer	15	\$44	\$660
1.2	SRS: Testing & Accessibility Sections	SRS (Team B)	M1	Business Analyst	15	\$49	\$735
1.3	Testing Baseline Audit	Testing Baseline Document	M1	QA / Tester	20	\$49	\$980
1.4	Test Plan Creation	Test Plan	M1	QA / Tester	20	\$49	\$980
1.5	Coverage Improvement Plan	Coverage Roadmap	M1	QA / Tester	15	\$49	\$735
1.6	Accessibility & VPAT Planning	VPAT / Accessibility Plan	M1	QA / Accessibility Tester	30	\$49	\$1470
1.7	Environment Setup & App Validation	Running App Screenshot	M1	Software developer	5	\$64	\$320
2.1	Updated Project Plan & SRS	Updated Docs (Tracked Changes)	M2	Technical Writer	15	\$44	\$660
2.2	Technical Design / Technical Architecture	Technical Design Documentation	M2	Technical Architect	50	\$63	\$3150
2.3	Software Test Plan (Detailed)	Software Test Plan	M2	QA / Tester	40	\$49	\$1960
2.4	Initial Unit Tests	Unit Test Implementation	M2	Software Developer	40	\$64	\$2560
2.5	Initial E2E Tests	E2E Test Implementation	M2	QA / Tester	40	\$49	\$1960

2.6	Early Accessibility Review	Accessibility Findings	M2	QA / Accessibility Tester	10	\$49	\$490
3.1	Expanded Test Coverage	95% Coverage Progress	M3	Software Developer	30	\$64	\$1920
3.2	Regression Testing After Integration	Regression Test Results	M3	QA / Tester	30	\$49	\$1470
3.3	Programmer Guide	Programmer Guide	M3	Technical Writer	10	\$44	\$440
3.4	Accessibility & 508 Review	Updated Accessibility Evidence	M3	QA / Accessibility	15	\$49	\$735
3.5	VPAT Draft	VPAT Draft	M3	Technical Writer	40	\$44	\$1760
4.1	Final Test Coverage Report	95% + Verified Coverage	M4	QA Tester	40	\$49	\$1960
4.2	Final Accessibility Evidence	Accessibility Documentation	M4	QA / Accessibility Tester	15	\$49	\$735
4.3	Completed VPAT	Final VPAT	M4	Technical Writer	40	\$44	\$1760
4.4	User Guide / Testing Sections	User Guide	M4	Technical Writer	30	\$44	\$1320
4.5	Test Report	Final Test Report	M4	QA / Tester	40	\$49	\$1960
4.6	Final Quality Review & Demo	Demo Evidence	M4	QA / Tester	10	\$49	\$490
Totals					655		33130

Budget Summary Table

The budget summary below consolidates all WBS line items by role category, providing a high-level view of total estimated hours and costs across the full project lifecycle:

Budget Role Category	Hourly Rate	Estimated Hours	Estimated Cost
Team Lead / PM	\$48	40	\$1920
Business Analyst/Requirements Analyst	\$49	15	\$735
UI/UX Designer	\$47	N/A	N/A
Technical Architect / DevOps / Cloud Architect	\$63	50	\$3150
Software Engineer / Developer / Programmer	\$64	75	\$4800
QA / Tester / Accessibility Tester	\$49	325	\$15925
Technical Writer / Documentation Specialist	\$44	150	\$6600
Total		655	\$33,130

EVM Reporting

Earned Value Management (EVM) is used to track Team B's progress against the planned budget and schedule. The table below is updated at each milestone to reflect actual hours worked, percent completion, earned value, and overall task status:

WBS ID	Task	Role Category	Planned Hours	Actual Hours	Percent Complete	Planned Value	Earned Value	Actual Cost	Status
1.0	Team Lead Coordination & Milestone Management	Team Lead / PM	40	10	25%	\$1920	\$480	\$480	Green
1.1	Project Plan Development	Technical Writer	15	20	100%	\$660	\$660	\$880	Green
1.2	SRS: Testing & Accessibility Sections	Business Analyst	15	7	100%	\$735	\$735	\$343	Green
1.3	Testing Baseline Audit	QA / Tester	20	25	100%	\$980	\$980	\$1225	Green
1.4	Test Plan Creation	QA / Tester	20	13	100%	\$980	\$980	\$637	Green
1.5	Coverage Improvement Plan	QA / Tester	15	15	100%	\$735	\$735	\$735	Green
1.6	Accessibility & VPAT Planning	QA / Accessibility Tester	30	0	0%	\$1470	0	0	Red
1.7	Environment Setup & App Validation	Software developer	5	0	0%	\$320	0	0	Red
2.1	Updated Project Plan & SRS	Technical Writer	15	0	0%	\$660	0	0	Red

2.2	Technical Design / Technical Architecture	Technical Architect	50	0	0%	\$3150	0	0	Red
2.3	Software Test Plan (Detailed)	QA / Tester	40	0	0%	\$1960	0	0	Red
2.4	Initial Unit Tests	Software Developer	40	0	0%	\$2560	0	0	Red
2.5	Initial E2E Tests	QA / Tester	40	0	0%	\$1960	0	0	Red
2.6	Early Accessibility Review	QA / Accessibility Tester	10	0	0%	\$490	0	0	Red
3.1	Expanded Test Coverage	Software Developer	30	0	0%	\$1920	0	0	Red
3.2	Regression Testing After Integration	QA / Tester	30	0	0%	\$1470	0	0	Red
3.3	Programmer Guide	Technical Writer	10	0	0%	\$440	0	0	Red
3.4	Accessibility & 508 Review	QA / Accessibility	15	0	0%	\$735	0	0	Red
3.5	VPAT Draft	Technical Writer	40	0	0%	\$1760	0	0	Red

4.1	Final Test Coverage Report	QA Tester	40	0	0%	\$1960	0	0	Red
4.2	Final Accessibility Evidence	QA / Accessibility Tester	15	0	0%	\$735	0	0	Red
4.3	Completed VPAT	Technical Writer	40	0	0%	\$1760	0	0	Red
4.4	User Guide / Testing Sections	Technical Writer	30	0	0%	\$1320	0	0	Red
4.5	Test Report	QA / Tester	40	0	0%	\$1960	0	0	Red
4.6	Final Quality Review & Demo	QA / Tester	10	0	0%	\$490	0	0	Red

8. Risk Management

Risk	Likelihood	Impact	Mitigation
Team member unavailable / unresponsive	Medium	High	Weekly peer evaluations; escalate to professor if unresponsive after 48hrs
Dev environment setup delays	High	Medium	Alex provides setup support; all members target env setup by May 25
Merge conflicts from other teams	High	Medium	Frequent staggered merges; pull from team-b-develop daily before new work
95% coverage target not achievable	Medium	High	Begin coverage audit in Milestone 1; identify gaps early and prioritize

VPAT format/scope unclear	Medium	Medium	Raise with Professor Assadullah and Roy by end of Week 2
661 UI components delayed or non-compliant	Medium	Medium	Validate components as they arrive; document findings; communicate early
Missing or misconfigured environment variables/API keys	High	High	Maintain shared environment variable documentation and validate required configuration before integration testing
CI/CD pipeline instability during integration testing	Medium	Medium	Require pull request validation checks and automated test execution before merges to team-b-develop

8. Open Questions

The following table documents questions that required resolution with Professor Assadullah, Roy, or Joriel Rivas prior to or during Milestone 1. Items marked Resolved have been confirmed and incorporated into the project plan. Items still open will be addressed no later than Milestone 2:

Question	Directed To	Status
What is the current code coverage %?	Repo audit	<i>Resolved</i>
What testing framework is already in place?	Repo audit / Joriel	<i>Resolved</i>
What VPAT format/level of completion is expected?	Professor / Roy	<i>Resolved - Full conformance statement by M4</i>
What workflows are highest priority for E2E testing?	Professor / Roy	<i>Resolved - Flexibility given; all complete by M3</i>
What accessibility tools should be used (axe, WAVE, etc.)?	Professor / 661	<i>Resolved - axe core confirmed</i>
What hourly rates should be used for cost estimation?	Joriel / Roy	<i>Resolved - BLS rates per class standard</i>
What level of support will SWEN 661 provide for VPAT?	Professor / 661	<i>Resolved - SWEN 661 can provide initial support only if Prof. Minagar agrees, but responsibility and accountability lies with Team B.</i>
Should CI/CD be built via CloudFormation/IaC?	Professor	<i>Resolved — CloudFormation required; no AWS UI configuration</i>
What is the tiered coverage threshold for high-risk vs UI modules?	Professor	<i>Resolved — 100% transactional logic; 90% visual/placement</i>

9. Document Approval

By signing below, each team member confirms they have reviewed this Project Plan and accept their assigned responsibilities.

Name	Signature	Date
Alex Yom	Alex Yom	05/28/2026
Brice Tikum	Brice Tikum	05/26/2026
Prashanth Saseenthara	Prashanth Saseenthara	05/29/2026
Lekan Ogunsanya	Lekan Ogunsanya	05/27/2026